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IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-514

1. Topic (12 words max)

Hemora - Identification of Hematological Diseases using Deep Learning and Microscopic Blood Smear Image Processing

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Hematological diseases affect the blood and the organs responsible for producing blood. They are a big health problem in Sri Lanka and all over the world. Common hematological diseases like Leukemia, Thalassemia, Iron Deficiency Anemia (IDA), and Sickle Cell Disease can cause serious problems like organ damage, developmental delays, and death if not diagnosed early [1]. Traditional diagnostic approaches rely heavily on manual microscopic examination of peripheral blood smear slides. It is time-consuming, labor-intensive and observations can be different from one observer to the other [2].

The high rate of hematological disorders is especially concerning in Sri Lanka because of a number of unique causes. Thalassemia has one of the highest impact rates in the world, affecting around 2.2% of Sri Lankans [3]. Nearly 36% of women of reproductive age and 15% of children under five have iron deficiency anemia, which continues to be the most prevalent nutritional problem [4]. Over the past ten years, acute lymphoblastic leukemia has become much more common in children, and low survival rates are a result of delayed diagnosis [5].

There is a shortage of trained hematologists and laboratory technicians especially in rural areas in Sri Lanka. This will delay the diagnosis and treatment [6]. Additionally, misdiagnosis or late detection affect treatment outcomes and quality of life especially for genetic disorders like Thalassemia and Sickle Cell Disease [7].

The healthcare system in Sri Lanka faces many obstacles in delivering timely and precise hematological diagnostics. The limited availability of specialized devices like flow cytometers and automated cell counters in local hospitals demands the use of manual microscopy [8]. The unequal distribution of healthcare resources between urban and rural regions causes difference in

capabilities, causing many patients in rural regions to travel long distances for proper hematological tests [9]. In government hospitals, the current diagnostic processes often face delays ranging from 3 to 7 days for routine blood tests, which can be crucial in situations such as leukemia [10].

Deep learning and image processing has made it possible to create AI-powered systems that can automatically find defects in blood in peripheral smear images. Convolutional Neural Networks (CNNs) in particular, have shown excellent performance in image classification and feature extraction from medical images [11,12]. Deep learning models that have been trained on labeled datasets can automatically find blast cells in leukemia or hypochromic microcytic cells in IDA with a high level of accuracy within a very short time period [13,14].

References:

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[2] V. Sharma et al., "Manual vs. automated microscopy in anemia detection," *Journal of Hematology*, 2019.
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[4] T. Weatherall, "Thalassemia in Sri Lanka: A local overview," *Sri Lanka Journal of Child Health*, 2018.
[5] A. Rehman et al., "Classification of Acute Lymphoblastic Leukemia using Deep Learning," *IEEE Access*, vol. 7, pp. 142-149, 2019.
[6] M. Talo, "Automated classification of blood cells using CNN," *Computer Methods and Programs in Biomedicine*, 2021.
[7] G. Li et al., "A CNN-Based System for Detection of Sickle Cell Anemia in Blood Smears," *Health Informatics Journal*, 2020.
[8] D. Sarki et al., "Image Preprocessing in Classification and Identification of Diabetic Eye Diseases," *Data Science and Engineering*, 2021.
[9] K. Premawardhena et al., "Thalassemia in Sri Lanka: Implications for the future health burden," *PLoS Medicine*, vol. 2, no. 4, 2005.
[10] Ministry of Health Sri Lanka, "National Nutrition and Micronutrient Survey," Colombo: Medical Research Institute, 2019.
[11] R. A. S. Lakmal et al., "Laboratory infrastructure and diagnostic capabilities in Sri Lankan hospitals," *Ceylon Medical Journal*, vol. 66, no. 2, pp. 45-52, 2021.
[12] P. Jayawardana et al., "Healthcare accessibility in rural Sri Lanka: Challenges and solutions," *International Journal of Health Policy*, vol. 15, no. 3, pp. 78-85, 2022.
[13] S. M. Perera et al., "Diagnostic delays in government hospitals: A Sri Lankan perspective," *Journal of Health Management*, vol. 24, no. 1, pp. 23-31, 2022.
[14] A. Fernando et al., "Economic impact of hematological diseases in developing countries: The Sri Lankan experience," *Health Economics Review*, vol. 11, no. 1, pp. 1-12, 2021.

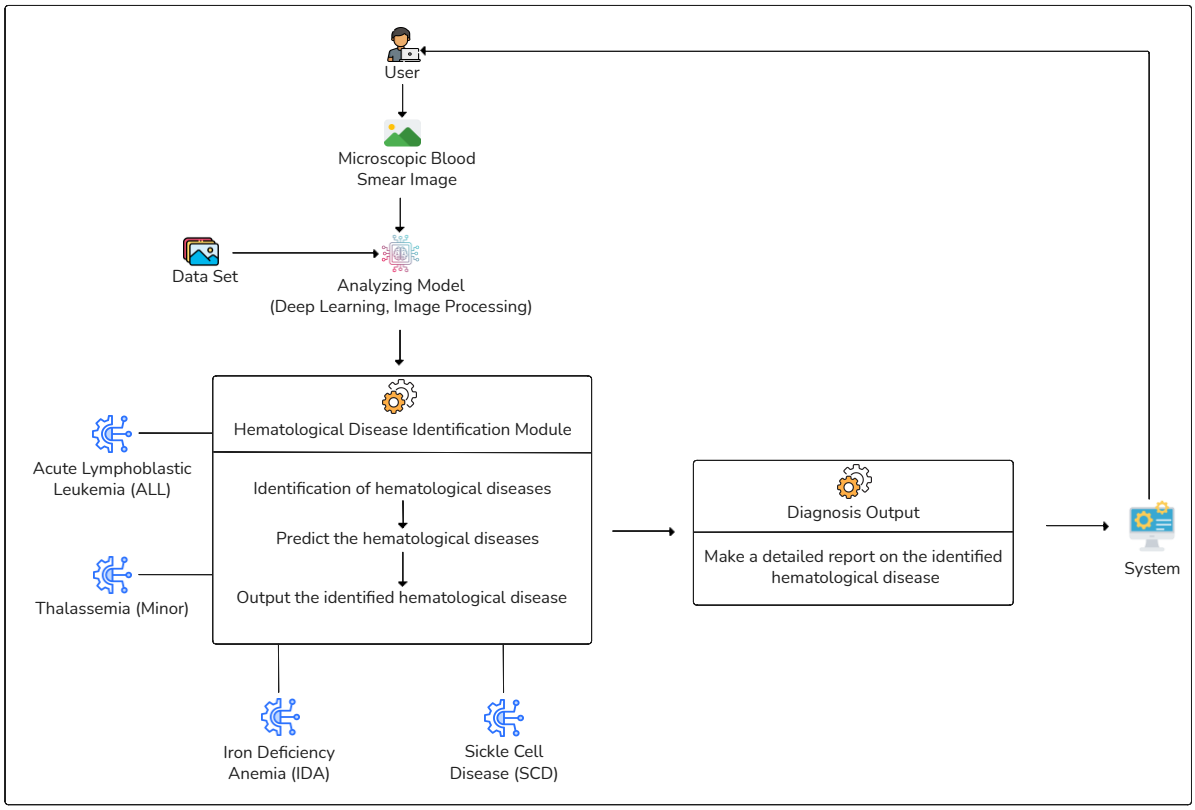
6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Hemora: Four common hematological diseases are the focus of this research. Serious health risks are posed by these diseases, particularly in rural areas where access to prompt and accurate medical diagnosis is limited. To address these challenges, a deep learning-based solution is being proposed using image processing techniques. A web-based solution is planned to be implemented for this problem.

The dataset is mainly contained with blood smear images, which clearly illustrate the microscopic views of blood cells. Relevant features will be efficiently extracted and learned by Convolutional Neural Networks (CNNs), YOLO, Vision Transformers (ViTs), or a Hybrid method. Effectiveness in image processing and object detection is widely recognized for these models due to their ability to identify meaningful patterns in blood smear images.

A labeled dataset of blood smear images will be used to represent each disease for training the model. During the training phase, subtle visual features that separate one disease from another will be learned by the selected model. After training is completed, separate datasets that were not used during training will be used for validation and testing. This process is carried out to evaluate the model's ability to generalize and to ensure its accuracy when presented with new images.

By applying this approach, it is expected that hematological diseases can be identified more efficiently, reducing diagnostic time and supporting medical professionals in making faster and more accurate decisions. This method is considered to have potential for future use in clinical systems.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Hematological diseases such as Leukemia (ALL), Thalassemia (Minor), Iron Deficiency Anemia (IDA), Sickle Cell Disease (SCD) aimed to be identified using deep learning-based image processing of blood smear images. This research project aims to enhance the patient outcomes and optimize disease management by developing models that are capable of early diagnosis.

Specialized expertise knowledge in the healthcare domain is essential for the success of this project. Medical image analysis requires the expertise in processing and interpreting the blood smear images using algorithms which are designed for medical applications. A deep understanding of hematology and blood cells architecture is required to effectively identify the abnormality in the blood cells that has affected. In addition, expertise of advanced machine learning techniques including the development of hybrid models which integrate multiple algorithms, methodologies are essential. The hybrid models will be fine-tuned with specific hyperparameters to achieve superior performance in disease detection.

Knowledge requires a comprehensive understanding of disease pathology, symptoms, abnormality visible in the blood smear images. Comprehensive awareness about the fundamentals of machine learning and deep learning is essential for training the effective hybrid models. Selection of appropriate algorithms, optimizing them through the hyperparameter tuning and integrating multiple approaches to enhance the accuracy and reliability. A thorough understanding of medical imaging principles and the ability to interpret complex visual data are also crucial.

Data requirements are stringent, demanding of a large, annotated datasets of blood smear images. The datasets should cover a wide range of hematological diseases to ensure the robustness and generalizability. High quality from diverse sources is essential for capturing the variability in disease presentation. Furthermore, the clinical acceptance of model outputs depends on their interpretability. Performance and the reliability of the model can be improved, especially for uncommon conditions by using strategies like multi-layered models, heatmaps for model predictions, and the creation of synthetic data to supplement training datasets.

In conclusion, Identification of Hematological Diseases Using Deep Learning and Microscopic Blood Smear Image Processing requires a combination of domain expertise, comprehensive knowledge of disease pathology and machine learning and the access to variety of high-quality data. By combining these elements, the goal of effectively identifying hematological disorders will be achieved through the creation of hybrid models and the advanced technologies.

8. Objectives and Novelty

Main Objective

This research is to develop an automated system for the identification of hematological diseases using microscopic blood smear image processing. The system will focus on detecting four specific diseases Acute Lymphoblastic Leukemia (ALL), Thalassemia (Minor), Iron Deficiency Anemia (IDA), and Sickle Cell Disease (SCD). By applying deep learning techniques, the proposed solution aims to improve diagnostic accuracy, reduce analysis time, and support medical personnel in low-resource clinical environments in Sri Lanka.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Aluthge A.D.M.S IT22371218	Detection and identification of Acute Lymphoblastic Leukemia (ALL)	• Gather a dataset of peripheral blood smear images and complete blood count (CBC) reports from suspected Acute Lymphoblastic Leukemia (ALL) cases. • Preprocess images by resizing, enhancing contrast, and segmenting cells using computer vision techniques. • Image segmentation to isolate white blood cells, focusing on lymphoblasts which are key indicators in ALL.	Developing a suitable model like Convolutional Neural Networks (CNN) with higher number of layers or YOLO or Vision Transformers (ViTs), or a Hybrid method for detection of Acute Lymphoblastic Leukemia (ALL) involves effectively identifying lymphoblasts and other cells. Unlike current methods that rely on a single type of diagnostic input, analyzing visual characteristics of lymphoblasts from microscopic images with CBC parameters like WBC count, lymphoblast percentage, and hemoglobin levels, a higher accuracy can be achieved.

		• Development of deep learning models after choosing appropriate model for classifying blast cells and other cells. • Integrated CBC data (e.g., WBC count, lymphoblast percentage) into a hybrid decision-making model. • Evaluation of model's accuracy and performance on locally gathered clinical datasets. • Validation of model using real patient data to ensure they work well in practical hospital situations. • Implementation of a user-friendly web-based interface for health care professionals to use easily without any difficulty.	This combination improves diagnostic reliability and lowers the chances of false positives or negatives. The system analyzes white blood cell (WBC) shapes in blood smear images to identify lymphoblasts - abnormal cells with large nuclei and distinctive nuclear features. It also examines CBC patterns like high white cell counts with immature cells, low platelets, and anemia. By combining image analysis with blood count data, the system can distinguish ALL from normal conditions Additionally, a system is developed as a web-based application for Sri Lankan laboratories to input peripheral blood smear images and complete blood count (CBC) reports and fasten the diagnosis and reduce human errors.
Disanayake R.G.O.S IT22368348	Detection and identification of Beta-Thalassemia (Minor)	• Collection of peripheral blood smear images and complete blood count (CBC) reports from suspected Beta-Thalassemia (Minor) cases.	This system introduces the first Sri Lankan-specific AI model that combines deep learning-based blood smear image analysis with

		• Preprocess images by resizing, enhancing contrast, and segmenting cells using computer vision techniques. • Development of deep learning models after choosing appropriate model for classifying morphological features such as microcytosis, hypochromia, and target cells. • Integration of CBC parameters (MCV, MCH, RBC count) into a hybrid classification framework using machine learning techniques. • Evaluation of model accuracy and performance on locally sourced clinical datasets. • Validation of model using real patients' data to ensure they work well in practical hospital situations • Implementation of a user-friendly web-based interface for health care professionals to use easily without any difficulty.	Complete Blood Count (CBC). The hybrid model uses deep learning techniques to classify red blood cell morphologies such as microcytic, hypochromic, and target cells. It also incorporates CBC data like MCV, MCH, and RBC count using appropriate machine learning algorithms. Existing identification systems often have issues such as low generalizability, limited feature coverage, and reliance on only one type of input data. In contrast, the proposed hybrid model combines both visual and numerical diagnostic features, allowing for more accurate and reliable detection of Beta-Thalassemia (Minor). It is especially useful in differentiating thalassemia from other conditions with similar blood profiles, like Iron Deficiency Anemia (IDA). The system works by learning detailed differences in red blood cell (RBC) shapes from blood smear

			images. It focuses on identifying features like target cells and severe hypochromia, which are more commonly seen in thalassemia. Along with this, the system also looks at specific CBC patterns such as low MCV and MCH, but with a normal or high RBC count which are typical signs of Beta-Thalassemia (Minor) and not usually found in Iron Deficiency Anemia (IDA). By combining both image features and CBC data, the system can accurately tell the difference between diseases that look similar, helping to reduce wrong diagnoses. The system is developed as a web-based platform, providing remote access for laboratory staff and healthcare professionals. It supports easy data upload, fast processing, and smooth integration into clinical workflows in hospitals and diagnostic centers across Sri Lanka.

Dassanayake E.G.N.L IT22361936	Detection and identification of Sickle Cell Disease (SCD)	• Collection of peripheral blood smear images and complete blood count (CBC) reports from Sickle Cell Disease (SCD) cases. • Segment red blood cells (RBC) from the background using computer vision techniques to isolate individual cells for efficient analysis. • Develop a deep learning model by selecting the most suitable machine learning algorithm to identify abnormal RBC shapes, especially focusing on detecting sickle shaped cells and related morphological signs. • Integrate Complete Blood Count (CBC) parameters such as Hemoglobin, MCV and RBC count into the hybrid machine learning framework. • Train the hybrid model using the collected clinical datasets (blood smear images) to evaluate the accuracy and the performance.	This introduce a hybrid deep learning model for detecting Sickle Cell Disease (SCD) which is a genetically inherited disease in Sri Lanka. The system combining blood smear image analysis with CBC data such as Hemoglobin, MCV and RBC count to improve the diagnosis accuracy. Sickle Cell Disease (SCD) is caused by a mutation in the gene which increases the abnormal hemoglobin (HbS) level, which causes the red blood cells (RBC) to convert into sickle shape under low oxygen conditions. The model integrates R-CNN and YOLO into a hybrid model, which improves both the accuracy and efficiency of identification, even in difficult conditions in the images. This approach will address defects such as human errors and time consuming in the manual approach which is currently in use. Key CBC parameters (like Hemoglobin, MCV, and RBC count) are incorporated to support

		• Validate the model results with real patient data to ensure it works properly in the clinical labs and the hospitals. • Implementation of a user-friendly web-based interface for health care professionals to use easily without any difficulty.	decision-making, enhancing diagnostic reliability. Combination of these elements will yield the solution as a web-based application facilitate the laboratory staff and the healthcare professionals.
M.A.S Sachinthanee IT22326140	Detection and identification of Iron Deficiency Anemia (IDA)	• Collect peripheral blood smear (PBS) images, complete blood count (CBC) reports from patients suspected of Iron Deficiency Anemia. • Preprocess peripheral blood smear images by enhancing contrast, normalizing color, and segmenting red blood cells using computer vision techniques to highlight morphological features like microcytosis, hypochromia, anisocytosis, and poikilocytosis. • Select and develop deep learning models to classify morphological features in peripheral blood smear images indicative of IDA, such as	This system represents a significant advancement in diagnosing Iron Deficiency Anemia (IDA) in Sri Lanka by integrating deep learning analysis of blood smear images with numerical data from Complete Blood Counts (CBC).By combining these two supporting information sources, the system can identify subtle morphological changes in red blood cells such as microcytosis(small),shape of the cell and hypochromia(pale), while simultaneously understanding key hematological parameters like MCV, MCH, and RBC count. The system works by analyzing detailed differences in red blood cell

		small, pale red blood cells and shape variations. • Extract key CBC parameters (e.g., hemoglobin, MCV, MCH) and integrate these numeric features with image derived features into a hybrid machine learning framework for improved diagnostic accuracy. • Use local patient data to train the system and test how well it detects iron deficiency anemia by checking how accurate and reliable it is. • Validate the model using real patient data from hospitals to confirm practical applicability. • Develop an efficient and user-friendly, web-based interface to identify IDA.	(RBC) shapes from blood smear images, focusing on features like microcytosis, hypochromia, anisocytosis, and pencil-shaped cells, which are typical in iron deficiency anemia (IDA). It also examines CBC patterns such as low mean corpuscular volume (MCV) and mean corpuscular hemoglobin (MCH) combined with a low RBC count, which help distinguish IDA from other conditions. By combining blood smear features with CBC data, the system accurately identifies IDA and reduces the chances of misdiagnosis. This system overcomes the limitations of current methods by accurately distinguishing IDA from similar conditions like thalassemia. The model is specifically trained on data relevant to the Sri Lankan people, improving its diagnostic sensitivity and specificity in local clinical contexts. Additionally, the software is also user friendly, with a web-based interface that allows

			laboratory staff and doctors upload data remotely, get immediate responses.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Aluthge A.D.M.S IT22371218	This component focuses on creating a system to detect Acute Lymphoblastic Leukemia (ALL) using blood smear images and CBC data. It connects with Software Engineering through modular design, pipeline integration, testing the system using real clinical data and deployment. Also, a web-based solution will be created that will enable users to enter peripheral blood smear images and CBC reports and obtain diagnostic results. The system will be implemented in Sri Lankan labs to fasten the clinical procedures. Important SE practices like scalability, error handling, and user-centered interface design and also techniques like hybrid deep learning methods are applied.
Disanayake R.G.O.S IT22368348	This component uses core software engineering concepts to design and develop a web-based disease detection system. It includes implementing deep learning models, creating data preprocessing pipelines, and building a hybrid prediction engine that uses both image data and CBC parameters. The work involves full-stack development such as integrating the model, connecting the frontend and backend, and testing the system using real clinical datasets. It applies advanced software design, machine learning techniques, and system deployment suitable for healthcare applications.
Dassanayake E.G.N.L IT22361936	The component of identifying Sickle Cell Disease (SCD) by using the hybrid deep learning models will be aligned with the software engineering principles, following the software development lifecycle (SDLC). Security of the data (blood smear images) that has taken is highly ensured using the software best practices. High performance of the functionalities aims to deliver efficient output to the end users. Integration of machine learning and deep learning will be successfully achieved in the web-based application, while maintaining the system reliability.
M.A.S Sachinthanee IT22326140	This component plans to develop a diagnostic system for Iron Deficiency Anemia (IDA) detection by combining blood smear image analysis with Complete Blood Count (CBC) data. And it follows software engineering principles with modular design, pipeline integration and thorough tests to ensure reliability. It features error handling to address data inconsistencies and a user-centered web interface, enhancing usability and maintainability which is more important in software engineering. These practices make the system scalable, strong, and well-suited for accurate Iron Deficiency Anemia detection in clinical environments.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Dinuka	Wijendra	23/06/2025
Co-Supervisor	Ms.	Jenny	Krishara	23/06/2025
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.